Preventing the Next Pandemic: Epidemiological Compartmental Models, the Reproduction Number, and Determining Intervention Strategies

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Abstract

After the COVID-19 pandemic upended daily life in 2020, questions about how disease control agencies around the world could better prepare for the spread diseases arose. Epidemiology is the study of the spread of diseases, and in this paper, we explore how epidemiological models can inform disease intervention strategies. In particular, we show how epidemiological systems of ordinary differential equations (ODEs) can project the long-term trajectories of diseases, along with how along small changes in assumptions can result in large impacts on the model. Additionally, we discover the importance of the reproduction number $\mathcal{R}_0$ in determining long-term behavior of a disease while also exploring how models can be manipulated to uncover key insights that inform disease intervention, such as designing quarantining policy or setting vaccination thresholds. Studying the spread of disease through these models and methods can contribute to disease mitigation in the long-run.

Keywords: Epidemiology, SIR Model, Ordinary Differential Equations

1. Introduction

Epidemiological modeling is a fascinating and vital branch of academia today. Past pandemics, such as COVID-19, upended daily life and changed the ways that societies respond to the endemic spread of diseases. Since the pandemic, scholars have used vast data collections to help inform disease control authorities in optimizing their response strategies, such as implementing social distancing measures [1]. Continuing the study of epidemiological modeling is necessary to help prevent and mitigate the effects of future pandemics.

The United States Centers for Disease Control (CDC) defines an epidemic as the "occurrence in a community or region of cases of an illness, specific health-related behavior, or other health-related event clearly in excess of normal expectancy [2]." Epidemics differ from pandemics in size; pandemics tend to cover a larger area, like a country or continent, while epidemics are confined to smaller regions. We define an infection as the presence of an illness in an individual, which—depending on the disease—can be spread between individuals. Moreover, infections are not indefinite; there is usually a finite infectious period where the individual has the infection and can pass it onto another individual, before ultimately losing the infection. Epidemiological models use these basic states and properties of infections to project the trajectory of a disease.

The core models of epidemiology tend to comprise compartmental systems of ordinary differential equations that map the flows between different segments of a population. One of the most basic—but foundational—models is the SIR [3]. The standard SIR model breaks up a given population into three different clusters based on phases of a disease flowing through that population: Susceptible (S), Infected (I), and Recovered (R). However, while SIR does model a disease's lifespan in a population, it rests on many assumptions that limit it. This leads to variations and expansions of these compartmental models, some of which can contain up to 8 compartments, or contain a small but complex change to the basic structure. These more complicated models often relieve some of the limitations of SIR and make it more realistic.

In this paper, we will show these variations between compartmental epidemiological models, along with their core result, the reproduction number, to answer the question: **How can public health agencies use epidemiological modeling to determine effective disease intervention strategies?** In the Methods section (2), we introduce a variety of core epidemiological models, building up from the standard SIR model and into more complex variations. In the section entitled "The Reproduction Number,

$\mathcal{R}_0$ (3)," we introduce the importance of the reproduction number, analyze long-term behavior of each model by its $\mathcal{R}_0$, and introduce a tool for calculating it. In the section entitled "Disease Intervention Analysis (4)," we derive conditions that dictate disease intervention from $\mathcal{R}_0$ by exploring the mathematical impact of practices like vaccination, quarantining, medication, and hospitalization.

2. Methods

In this section, we introduce the standard SIR model and its variants. SEIR, SEIQR, SEIHR and other variations are mentioned in more detail in the full paper.

2.1. Basic Epidemiological Models

We introduce the basic epidemiological models, which are foundational to modeling diseases, but are typically limited by assumptions.

2.1.1 Standard SIR Model

SIR models the flow of individuals between three compartments: Susceptible, Infected, and Recovered. This can be seen in Figure 1.

Because this is a compartmental model tracking inflows and outflows, the system of differential equations looks like

$$\begin{aligned} \frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I, \end{aligned} \tag{1}$$

where β represents the rate of susceptible individuals that become infected (per unit time), which is person-to-person, and $\gamma > 0$ represents the rate at which infected individuals recover (per unit time) such that $\frac{1}{\gamma}$ represents the average recovery period. The initial conditions are given by

$$\begin{aligned} &S(0) = S_0 \\ &I(0) > 0 \text{ such that } I(0) \ll S(0) \\ &R(0) = 0. \end{aligned} \tag{2}$$

The flow from S to I can be thought of as the product of contacts per day, the probability a contact is infected, and transmission rate. The flow from I to R is directly represented by the recovery rate, γ, scaled by the number of infectious individuals I. The dependent variables here are the sizes of each respective population cluster, $S(t)$, $I(t)$, and $R(t)$.

The parameters and their units are further explained in Table 1. To see the dynamics of the SIR model, visit the Supplementary Information in the full paper.

2.1.2 Parameters

The following table is a list of the parameters used in each basic model studied in this paper, accompanied by a brief explanation. For all rates, observe that the units are the proportion of the population per unit time.

Parameter	Explanation
N	Population
β	Rate of susceptible individuals becoming infected/exposed (Transmission)
γ	Rate of infected individuals becoming recovered
κ	Rate of exposed individuals becoming infected (Latency)
δ	Rate of infected individuals quarantining themselves
ξ	Rate of quarantined individuals that recover
α	Rate of infected individuals that become hospitalized
μ	Rate of hospitalized individuals that recover

Table 1: Epidemiological Model Parameters

2.1.3 Compartmental Flow

Figure 1 shows a multilayered flow chart including the compartments for the models of SIR, SEIR, SEIQR, and SEIHR. Please observe the legend to view the corresponding colors. Each flow contains parameters mentioned in Table 1.

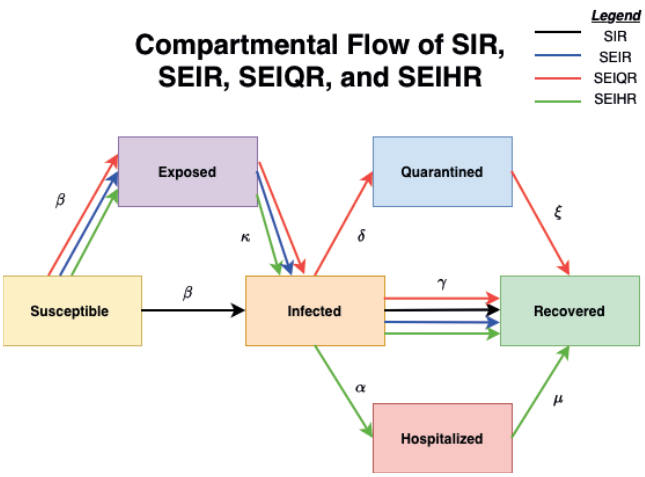


Figure 1: A compartmental flow map of the basic SIR, SEIR, SEIQR, and SEIHR models. Each model has its own arrow with a corresponding color and flow rate.

2.2. Epidemiological Models With Vital Dynamics

One of the simpler extensions to any epidemiological model is the addition of vital dynamics. This means that the model now accounts for births and deaths, allowing the dynamics of epidemiological models to more closely model reality. We introduce the SIR model with simple demography as an example.

2.2.1 SIR Model with Simple Demography

The SIR model with simple demography is defined as

$$\begin{aligned} \frac{dS}{dt} &= A - \beta SI - dS \\ \frac{dI}{dt} &= \beta SI - (d+\gamma)I \\ \frac{dR}{dt} &= \beta SI - dR, \end{aligned} \tag{3}$$

where the parameters β and γ are the same as in the basic SIR model, and the new parameters $A > 0$ and $d > 0$ represent the simple birth and death rates, respectively.

2.2.2 Compartmental Flow

We present a multilayered flow chart of the vital dynamics models covered in this paper. Please see the legend to view the corresponding color.

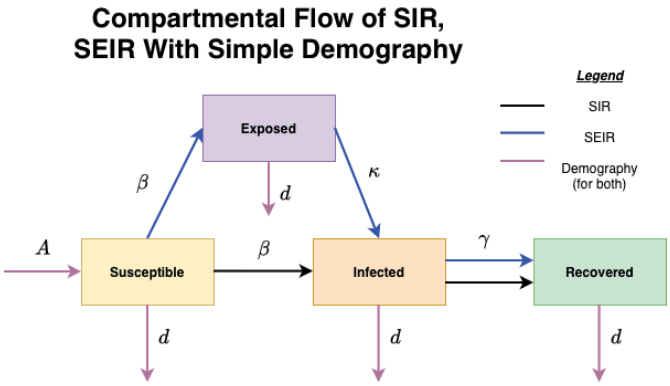


Figure 2: A compartmental flow chart of SIR and SEIR with simple demography. The parameters are the same as in Table 1, and the new parameters A and d are included to represent births and deaths from the population.

2.3. Models With Reinfection

Another common complication of epidemiological models is reinfection, which means that after being infected and recovering, individuals can once again return to the susceptible class. We introduce two reinfection models, SIRS and SEIRS, in this subsection.

2.3.1 SIRS

The SIRS model is defined as

$$\begin{aligned} \frac{dS}{dt} &= -\beta SI + \tau R \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I - \tau R, \end{aligned} \tag{4}$$

where the parameters β and γ are the same as in Equation (1), and the new parameter τ represents the rate at which recovered individuals return to the susceptible class, meaning that they lose immunity at rate τ [4].

2.3.2 Compartmental Flow of SIRS

We present a multilayered flow chart of the reinfection models covered in this paper: SIRS and SEIRS. Please see the legend to view the corresponding color.

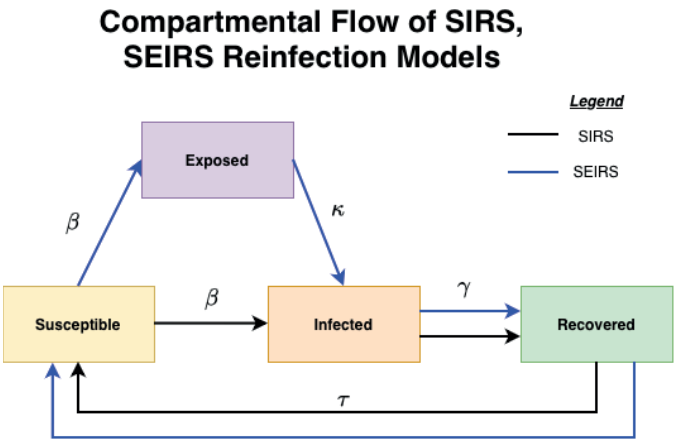


Figure 3: A compartmental flow chart of SIRS and SEIRS. The parameters are the same as in Table 1, and the new parameter τ is also included to represent the immunity loss rate.

3. The Reproduction Number, $\mathcal{R}_0$

In epidemiological models, a core result and tool for analysis is the basic reproduction number, $\mathcal{R}_0$ [5]. It represents the *average number of infections that are created by an infected individual in a given susceptible population.* In this section, we show how it models long-term behavior, display the different reproduction number formulas for the models in this paper, and introduce a method to help calculate reproduction numbers efficiently.

3.1. Steady State Analysis and the Reproduction Number

We begin our introduction to $\mathcal{R}_0$ by observing how steady states can lead to the definition of the reproduction number, followed by how to interpret $\mathcal{R}_0$ and how to calculate it.

3.1.1 Epidemiology Steady States: DFE and EE

To analyze long-term behavior of compartmental models, it is essential to understand how steady states are computed, and how they can inform the reproduction number

of a model.

Calculating a steady state can be done by setting each equation in the system equal to zero and solving for the independent variables. In the epidemiological literature, the equilibrium solution for models where $I = 0$ is known as the **Disease-Free Equilibrium (DFE)**, meaning the disease and the corresponding number of infections dies out. On the other hand, more complicated models can lead to the equilibrium solution where $I \neq 0$, meaning the model goes to an **Endemic Equilibrium (EE)** where the number of infections doesn't go to 0 and the disease survives.

This is important to note as we discuss the reproduction number $\mathcal{R}_0$. While the exact formula can vary across models, it is always defined on the model's steady states.

3.1.2 The Steady States of SIR's Variations

Using the method mentioned above, we can determine the steady states for the basic SIR model, SIR with vital dynamics, and SIRS.

The work outlined in the Supplementary Information section shows that the only steady state of the standard SIR model is $I = 0$, giving the equilibrium solution

$$(S^*, I^*, R^*) = (S^*, 0, N - S^*). \tag{5}$$

This means that the standard SIR model only has a DFE as a steady state, meaning that the disease will die in the long-run. This makes sense, given that in SIR the Susceptible class monotonically decreases until there are no more individuals that can be infected.

In the SIR model with vital dynamics, the steady states become slightly more complicated. Because of the constant rates A and d in this model, the DFE steady state becomes

$$(S^*, I^*, R^*) = \left(\frac{A}{d}, 0, 0\right). \tag{6}$$

This makes sense, since the susceptible population in the long run with no disease will become the proportion of new individuals over deaths. However, the SIR model with vital dynamics also captures an EE steady state. This is described by

$$(S^*, I^*, R^*) = \left(\frac{d+\gamma}{\beta}, \frac{\beta A - d(\gamma+d)}{\beta(\gamma+d)}, \frac{\gamma}{d} \cdot \frac{\beta A - d(\gamma+d)}{\beta(\gamma+d)}\right). \tag{7}$$

As we can see in this equilibrium, the Infected compartment is nonzero, which makes sense given the endemic equilibrium where a disease is proliferating in a population.

Finally, in the SIRS model, there are also two steady states (DFE and EE): the DFE is defined by

$$(S^*, I^*, R^*) = (N, 0, 0), \tag{8}$$

and the EE equilibrium solution is given as

$$(S^*, I^*, R^*) = \left(\frac{\gamma}{\beta}, \frac{\tau(\beta N - \gamma)}{\beta(\tau+\gamma)}, \frac{\gamma(\beta N - \gamma)}{\beta(\tau+\gamma)}\right) \tag{9}$$

Again, these steady states make sense. In the DFE, the entire population is healthy and susceptible, but the disease dies out [4]. In the EE, the Infected compartment is again nonzero, indicating the survival of the disease.

3.1.3 The Reproduction Numbers of SIR

Recall that the reproduction number $\mathcal{R}_0$ is intuitively defined as the *average number of secondary cases produced by an infected individual in a susceptible population* [5]. We can use the steady states from the previous section to start defining the different reproduction numbers across these SIR variations. This section focuses on the more intuitive approach to finding the formula for these basic models, but Section 3.3 defines a more rigorous process for finding the reproduction numbers across different models.

In the standard SIR model, recall the DFE steady state in Equation (5). If we focus on finding where $\frac{dI}{dt} > 0$ at the beginning of the disease (where $t = 0$ and $S^* = S_0$), then we can see when the infections grow. Observe that $\frac{dI}{dt} = I(\beta S_0 - \gamma)$, and

$$I(\beta S_0 - \gamma) > 0 \implies \frac{\beta S_0}{\gamma} > 1.$$

As it turns out, the reproduction for the standard SIR model is defined as

$$\mathcal{R}_0 = \frac{\beta S_0}{\gamma}. \tag{10}$$

We can similarly find the reproduction numbers for SIR with vital dynamics and SIRS by following the same intuition of seeking the condition that $I^* > 0$, and that S^* in the EE is less than S^* in the DFE.

For SIR with vital dynamics, recall from Equation (3.1.2) that the DFE requires $S^* = \frac{A}{d}$ and the EE requires $S^* = \frac{d+\gamma}{\beta}$. Using the condition mentioned above and rearranging, we see that

$$\frac{\beta A}{d(d+\gamma)} > 1,$$

where $\frac{A}{d} = S^*$. This is similar to Van den Driessche's defintion of the reproduction number [5] for this model, which is defined as

$$\mathcal{R}_0 = \frac{\beta S_0}{(d+\gamma)}. \tag{11}$$

Using the same approach for SIRS, we recall from Equation (3.1.2) that the EE requires $S^* = \frac{\gamma}{\beta}$, and to reach $I^* > 0$ in the EE, we require $N > \frac{\gamma}{\beta}$, meaning that we get

$$\frac{N\beta}{\gamma} > 1.$$

Like with the other two models, it turns out that the reproduction number is defined as

$$\mathcal{R}_0 = \frac{\beta N}{\gamma}. \tag{12}$$

The reproduction numbers that have been defined here make sense in accordance with the intuitive definition introduced above. Recall that $\frac{\beta}{\gamma}$ represents the rate of infections happening during the average infection period, multiplied by the initial susceptible population, S_0. Whether adding in deaths or factoring in reinfection, the reproduction number formula is always a quantity made with respect to the DFE that represents how many secondary infections an infected individual will make in a susceptible population.

3.2. Interpreting $\mathcal{R}_0$

We have just shown in the previous section that the condition to reach a reproduction number is $\frac{dI}{dt} > 0$, which gives us

$$\mathcal{R}_0 > 1.$$

We can use the reproduction number to determine the long-term behavior of a disease [6]. In fact, we can say that $\mathcal{R}_0 < 1$ means that the number of infections will decrease to zero and $\mathcal{R}_0 > 1$ means that the number of infections will rapidly increase (and perhaps die off later, based on the model.)

We can use phase portraits to illustrate how $\mathcal{R}_0$ dictates the long-term behavior of a disease over time. Each phase portrait shown in this section will plot S against I, which shows how the trajectories of the disease will change and go to a steady state based on the reproduction number.

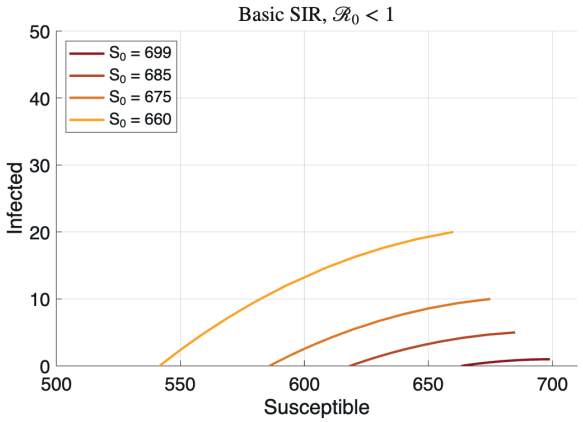


Figure 4: A phase portrait of the Standard SIR model with parameters $\beta = 0.0005$, $\gamma = 0.35$, and a set of $S_0 = \{699, 685, 675, 660\}$ as well as $I_0 = \{1, 5, 10, 20\}$, all leading to $\mathcal{R}_0 < 1$.

In Figure 4, we can see that, because of the low reproduction rate here, the infections rise slightly before quickly dying out. This shows that a low reproduction number can lead to quite rapid decreases in infections, causing the disease to die out. In the next figure, we observe how trajectories can change when $\mathcal{R}_0 > 1$.

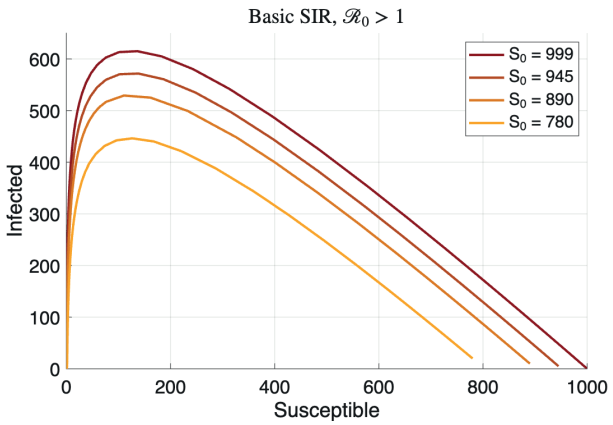


Figure 5: A phase portrait of the Standard SIR model with parameters $\beta = 0.4$, $\gamma = 0.1$, and a set of $S_0 = \{999, 945, 890, 780\}$ as well as $I_0 = \{1, 5, 10, 20\}$, all leading to $\mathcal{R}_0 > 1$.

In Figure 5, by reading the graph from right to left, we observe that as the disease grows, the number of infections grows, until it reaches a peak and drops to 0. The standard SIR, therefore, is a scenario where the disease always dies out, hence the existence of the DFE. This dynamic changes with the existence of vital dynamics.

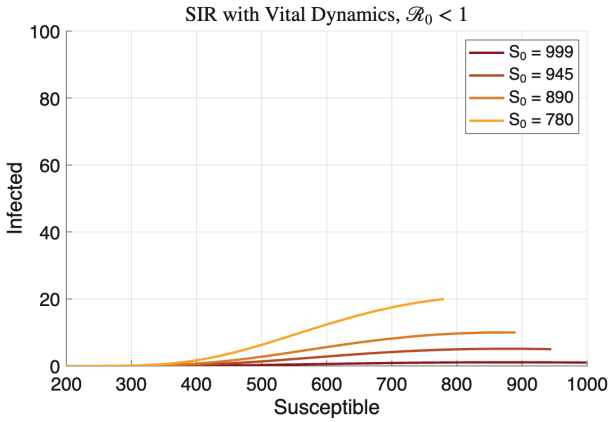


Figure 6: A phase portrait of the SIR model with vital dynamics and parameters $\beta = 0.0004$, $\gamma = 0.3$,$A = 10$, $d = 0.05$ and a set of $S_0 = \{999, 945, 890, 780\}$ as well as $I_0 = \{1, 5, 10, 20\}$ as well as $I_0 = \{1, 5, 10, 20\}$, all leading to $\mathcal{R}_0 < 1$.

We can see in Figure 6 that when $\mathcal{R}_0 < 1$, the disease dies out quickly again. Given the parameters, this makes sense; the relatively low transmission rate β coupled with the relatively high recovery rate γ means that out of the small number of infections that occur in this population, the infection period is so short that the impact is minimized.

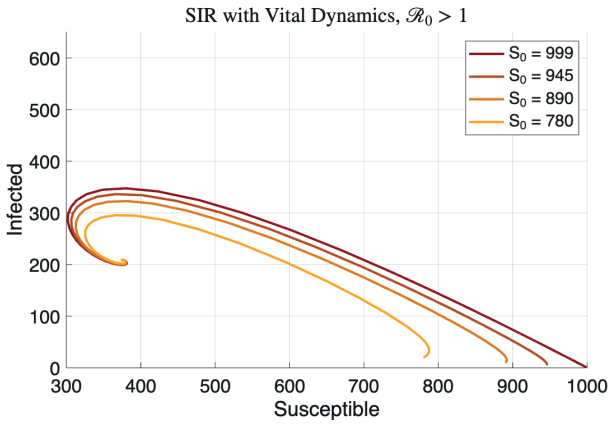


Figure 7: A phase portrait of the SIR model with vital dynamics and parameters $\beta = 0.0004$, $\gamma = 0.1$,$A = 50$, $d = 0.05$ and a set of $S_0 = \{999, 945, 890, 780\}$ as well as $I_0 = \{1, 5, 10, 20\}$, all leading to $\mathcal{R}_0 > 1$.

Figure 7 illustrates how the SIR model with vital dynamics interacts with the endemic equilibrium (EE) when $\mathcal{R}_0 > 1$. Unlike Figure 5, which showed the rise in infections in the basic SIR model that quickly dropped down to $I = 0$, this model spirals into a nonzero Infection class around $I = 200$. We can also observe similar dynamics with the SIRS model.

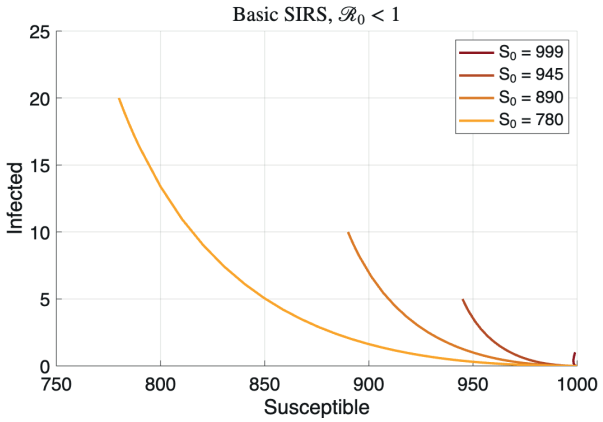


Figure 8: A phase portrait of the SIRS model with parameters $\beta = 0.0002$, $\gamma = 0.3$, $\tau = 0.05$, and a set of $S_0 = \{999, 945, 890, 780\}$ as well as $I_0 = \{1, 5, 10, 20\}$, all leading to $\mathcal{R}_0 < 1$.

This figure again illustrates how the infections quickly die out in a population where the reproduction number is less than 1. Even though reinfection is possible, the low immunity loss rate $\tau = 0.05$ makes the chances of reinfection slim, leading the infection class to stay quite low. We show this in contrast to the case of SIRS when $\mathcal{R} > 1$.

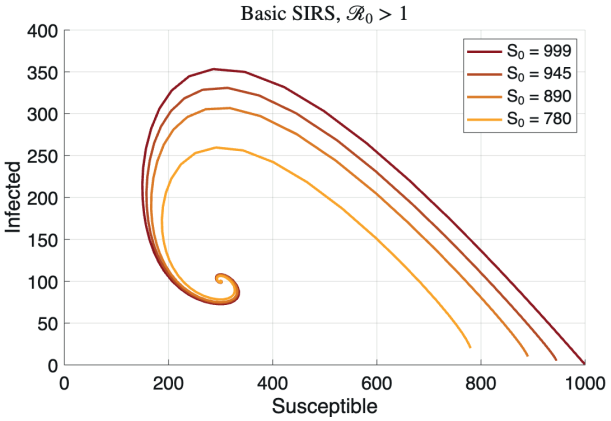


Figure 9: A phase portrait of the SIRS model with parameters $\beta = 0.001$, $\gamma = 0.3$, $\tau = 0.05$, and a set of $S_0 = \{999, 945, 890, 780\}$ as well as $I_0 = \{1, 5, 10, 20\}$, all leading to $\mathcal{R}_0 > 1$.

Figure 9 illustrates the opposite case to Figure 8. Here, the higher transmission rate β makes the disease easier to spread, resulting in a higher reproduction number. This is reflected in the spiral nature of the phase portrait (similar to Figure 7), which means the disease reaches a point in time where the infections never go to $I = 0$ and it survives in the population.

For more phase portraits of the different models covered in this paper, refer to the Supplementary Information. In

the next section, we describe the calculation process for rigorously defining the reproduction number, which can be useful as models grow in their number of compartments and complexity.

3.3. Calculating $\mathcal{R}_0$ With the Next Generation Matrix Method

As the reproduction number is derived from these epidemiological models—systems of ODEs—we can also infer long-term behavior more rigorously by introducing the concept of stability. From the plots in the previous subsection, the phase portraits illustrate the stability of steady states; in fact, when $\mathcal{R}_0 < 1$ the DFE is locally asymptotically stable, and when $\mathcal{R}_0 > 1$ the DFE becomes unstable. In models with vital dynamics where the EE exists, the EE becomes locally asymptotically stable for $\mathcal{R}_0 > 1$ [6].

In ODEs, a standard approach to performing stability analysis comprises of finding the equilibria, computing the Jacobian around those points, and solving for eigenvalues from a characteristic equation to determine local stability. This with the epidemiological models studied in this paper, but it is not always the easiest, especially if the model is complex.

The Next Generation Matrix (NGM) Method is an important tool for deriving $\mathcal{R}_0$, especially in larger compartmental models, where stability analysis through a Jacobian of the system can be laborious and complicated. Derived by Van Den Driessche et al., the following diagram lays out the process of finding the NGM, with the full process outlined in the full paper [6].

As stated previously, $\mathcal{R}_0$ determines stability. Van Den Driessche's Theorem states:

Theorem 3.1. *If x_0 is a DFE of the system $\frac{dx_i}{dt} = \mathcal{F}_i(x) - \mathcal{V}_i(x)$, then x_0 is locally asymptotically stable if $\mathcal{R}_0 = \rho(FV^{-1}) < 1$, and is unstable if $\rho(FV^{-1}) > 1$.*

3.3.1 Testing NGM Method on the Basic SEIR

We can verify the NGM method by testing it on the basic SEIR model. We can start by defining the DFE. Like the basic SIR, the DFE exists and is

$$(S^*, E^*, I^*, R^*) = (S^*, 0, 0, N - S^*).$$

Since there are two Infection compartments in this system, E and I, we can define $\mathcal{F}$ and $\mathcal{V}$ as

$$\mathcal{F} = \begin{pmatrix} \beta SI \\ 0 \end{pmatrix} \quad \& \quad \mathcal{V} = \begin{pmatrix} \kappa E \\ -\kappa E + \gamma I \end{pmatrix}.$$

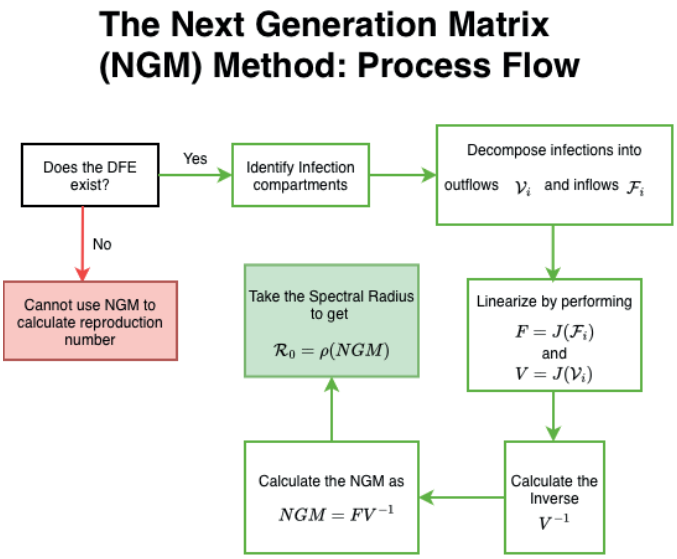


Figure 10: A flow chart that lays out the NGM Method for calculating $\mathcal{R}_0$.

After performing the linearization about the DFE to find F and V (the Jacobians of $\mathcal{F}$ and $\mathcal{V}$) and calculating V^{-1}, we see that

$$FV^{-1} = \begin{pmatrix} \frac{\beta S_0}{\gamma} & \frac{\beta S_0}{\gamma} \\ 0 & 0 \end{pmatrix}.$$

To compute the spectral radius, we just need to find the largest eigenvalue, which is $\lambda_1 = \frac{\beta S_0}{\gamma}$. Therefore, the reproduction number for the basic SEIR model is defined as

$$\mathcal{R}_0 = \frac{\beta S_0}{\gamma}. \tag{13}$$

The NGM found the reproduction number. While the NGM method doesn't necessarily cut down on algebraic complexity, it does optimize *computational* complexity, especially compared to finding the reproduction number by performing a typical stability analysis of a system. Take the SEIQR model, for example. The NGM method still requires calculating Jacobians and eigenvalues, but on a smaller scale; if we decided to just take the Jacobian of the entire system, we would have a 5x5 matrix and a fifth degree polynomial for the characteristic equation, which would take more computational power. By focusing on the infection component and using the NGM method, we develop a rigorous and useful way to define $\mathcal{R}_0$.

The NGM method is a computationally superior method for rigorously calculating a reproduction number that lends itself to interpretation in long-term disease behavior and it useful in informing different strategies of disease intervention.

4. Disease Intervention Methods

For years, researchers have relied upon epidemiological models to determine strategies to prevent the spread of disease [7]. In this section, we explore how to manipulate these models and their corresponding reproduction numbers to derive conditions on their parameters. We explore how models can be used to inform disease intervention strategies like vaccination, quarantining, medication, and hospitalization.

4.1. Vaccination Strategies

One of the most basic intervention strategies that can be inferred from epidemiological models is the vaccination threshold. Since the models studied in this paper all include "Susceptible" compartments, one can make the assumption that if a susceptible person becomes vaccinated, then they are no longer susceptible. Edelstein-Keshet starts by defining the fraction of the immunized population as p, thereby giving the susceptible fraction of the population as $1-p$, meaning $S = N(1-p)$ [4]. Since the effective reproduction rate becomes $\mathcal{R}_0' = (1-p)\mathcal{R}_0$, and we seek $\mathcal{R}_0' < 1$ for the disease to die out, a few algebraic steps point out that the vaccination threshold needs to be

$$p^* > 1 - \frac{1}{\mathcal{R}_0}, \tag{14}$$

where p is a fraction of the population. Paired with real-world data on diseases, researchers can calculate reproduction numbers, leading to an approximation of the vaccination threshold number [4]. For instance, data collected from England and Wales between 1942 and 1950 on the Whooping Cough epidemic calculated $\mathcal{R}_0 = 17$, leading to an approximate vaccination threshold of 94%. On the other hand, data collected from the 2009 South Africa H1N1 outbreak estimated a reproduction number of $\mathcal{R}_0 = 1.33$, which translates to vaccination threshold of $p^* > .25$. As a result, $\mathcal{R}_0$ is vital to calculating the vaccination threshold of a population.

To further illustrate the dynamics of the vaccination threshold, we measure how the vaccination threshold changes as β increases across models in Figure 11. This figure compares the vaccination thresholds based on their $\mathcal{R}_0$ definitions as β changes. Observe how, when β is small, the vaccination threshold percentage p^* across models is extremely small, but as the transmission rate increases, so too does the vaccination threshold. Additionally, notice how the threshold rate differs between models. In models like SIRS and SIRV, the susceptible class is higher than other models, meaning the vaccination threshold needs to be continually high.

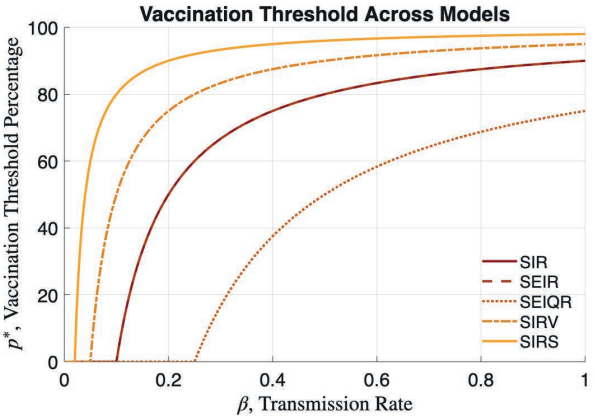


Figure 11: A plot comparing the transmission rate β to the vaccination threshold across the SIR, SEIR, SEIQR, SIR with vital dynamics (SIRV), and SIRS models. The parameters given are $\gamma = 0.1$, $\delta = 0.15$, $A = 10$, and $N = 5$.

4.2. The Impact of Medications

Another common way to fight illness is through medication. Generally, receiving some sort of medication can help speed up the recovery process (increasing γ in terms of epidemiological models) and cut down the potential infectious window in which infected individuals can spread a pathogen.

In the context of this paper, we assume that taking medication is mathematically equivalent to increasing γ, the recovery rate and that the act of taking medication will translate to a meaningful increase in it. To illustrate how different models interpret γ, we explore how the peaks of infections differ when γ is increased by taking medication.

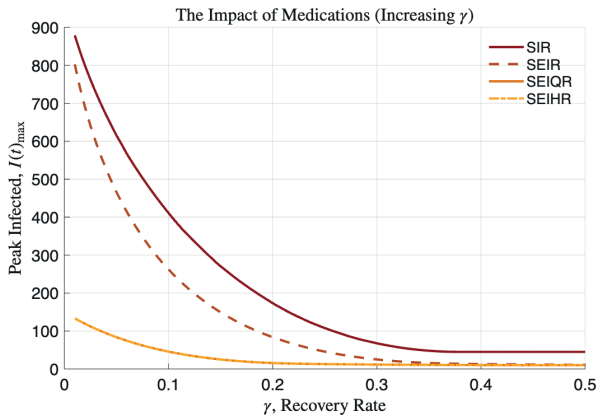


Figure 12: A plot of the relationship between the recovery rate γ, and the peak of infections, $I(t)_{max}$ across different models and using the parameters $\beta = 0.4$, $\gamma = 0.1$, $\kappa = 0.2$, $\delta = 0.15$, $\xi = 0.1$, and $\rho = 0.1$, and $S_0 = 950$.

We can see in Figure 12 how each model sees a decreasing $I(t)_{max}$ value (peak of infected class) as the recovery rate increases. This makes sense, given that the faster individuals are able to recover from an infection, the less time they spend in the Infection compartment, leading to a smaller peak in infections. Additionally, as the models become more complicated—moving from SIR to SEIR to SEIQR and SEIHR—the $I(t)_{max}$ value becomes more sensitive to lower recovery rates.

4.3. Adding the Hospitalization Dynamic

A key issue in fighting diseases is hospitalization. Infections, if strong enough, can sometimes cause symptoms to develop so severely that individuals may need to seek medical treatment in a hospital. This is the driving idea behind SEIHR [8].

In the SEIHR model (outlined further in the full paper), the α parameter represents the rate of infected individuals that become hospitalized. We test to observe where the peak hospitalization number, $H(t)_{max}$, occurs in order to find the critical hospitalization rate. One can observe that the peak of the hospitalized population peaks when $\alpha = 0.09$ and then flattens out afterwards. When paired with a hospital bed capacity number, like 100 in this case, public health agencies can determine the "sweet spot" hospitalization rates to avoid getting too close to hospital bed capacity.

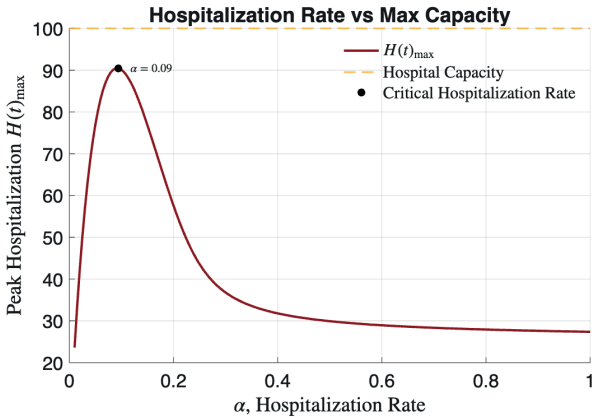


Figure 13: A figure of the relationship between α, the hospitalization rate and the peak of the Hospitalized population, $H(t)_{max}$ for the SEIHR model with parameters $\beta = 0.4$, $\gamma = 0.1$, $\kappa = 0.2$ and $\mu = 0.08$ and $S_0 = 950$.

5. Discussion

In this section we discuss results, limitations, and next steps for the analysis done in this paper.

5.1. Results

Throughout this paper, we've seen that epidemiological models like SIR and their variations are powerful tools to analyze and predict how a disease might spread in a given population. We've shown how models can become more complex by adjusting small assumptions, which can result in large impacts on the long-term behavior of a model. Moreover, the Disease-Free Equilibrium (DFE) and Endemic Equilibrium (EE) have been important in classifying systems, as well as deriving their reproduction number.

The reproduction number $\mathcal{R}_0$ is vital to modeling long-term behavior and deriving conditions on disease intervention. Not only can it be estimated by real datasets and interpreted quite simply, but it can determine the long-term behavior of a disease. By exploring how $\mathcal{R}_0$ can change across the models studies in this paper, we illustrated its versatility. Additionally, the NGM method is a rigorous method to calculate $\mathcal{R}_0$ and tie it back to the core mathematical concepts of stability. In addition to the power of the reproduction number, manipulating and rearranging the results of these models core models can uncover insights about slowing the spread of the disease, like determining what quarantine rates might be effective to quell a disease, or how medications might help slow down infections.

We can also draw qualitative conclusions about the models chosen. While SIRS might represent more realistic outcomes than SIR, there are certain diseases for which SIRS doesn't make sense and therefore SIR would be a better fit. For instance, once an individual has recovered from smallpox, they have lifelong immunity; in this case, SIRS is not as biologically useful as SIR. By providing several types of of models, this paper shows where some models perform better than others, and how that can also inform general disease intervention strategies.

5.2. Limitations & Next Steps

Despite all that we can infer from these models, they are still extremely limited in scope and relatively simple compared to other models, like SIDARTHE [9]. There are many assumptions and limitations baked into the models that we cannot completely trust the results of the models in a real-world context. Even when moving from the basic SIR model to a more complicated SIR with simple demography or the SIRS, the assumptions that those models rely upon still simplify real-world conditions. One such example is the lack of stochastic components in the models; these models are purely deterministic and don't take into account any randomness, which is inevitable with a disease.

Another one of the limitations of this paper is the lack of clean data. More advanced models have complex frameworks for estimating parameters, thereby giving their models and simulations a more enhanced sense of accuracy. However, even important parameters like β are extremely complicated to determine or estimate. Due to a lack of time, limited scope, and limited access to large datasets, we were not able to fully incorporate real data into our models. Doing so would give clearer and more accurate insight to the long-term behavior of a disease.

Finally, if there was more time in the semester, it would be interesting to tie in real disease intervention policy frameworks and analysis to compare and review the results of this paper. Since our research question focused on deriving disease intervention conditions, it would be interesting to study past policies and analyze them in the mathematical context of this paper.

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Jack Underwood



My name is Jack Underwood, and I'm a senior applied mathematics major at Middlebury College currently completing the Applied Math Modeling senior seminar. My path to mathematics was anything but linear (see what I did there?); growing up in Nashville, Tennessee, I was first in love with reading—particularly biographies. One book quickly reshaped my passions, however: *Surely You're Joking, Mr. Feynman*, the semi-autobiography of physicist Richard Feynman. It began an obsession with problem-solving that has never left. That obsession led me to join a local chess team, teach myself coding, and even a brief ambition to be a naval engineer—yet mathematics itself never captivated me. I was competent but uninterested and at times I resented math. That changed in AP Calculus AB, where a dedicated teacher and genuinely compelling problems finally sparked a deeper interest in the subject. I won math awards that year and arrived at Middlebury with a better sense of my interests.

Still, I wanted to explore academically during the beginning of my college career. As it turns out, that only lasted one semester. My first calculus course was the impetus for my decision to study math; from an exceptional professor, to genuinely interesting problems, and even the camaraderie formed in that class, I knew I wanted to declare as an applied math major.

Even after this discovery, the journey came with lows. During my sophomore year, I questioned whether I belonged. Coming from an under-resourced school district, advanced mathematics often felt like a discipline built for someone else. But I made a firm decision: as long as math compelled me, I would not quit. I would not let doubt—mine or anyone else's—override curiosity, and so I persevered.

As I wrap up my time here at Middlebury, I want to leave a message to my younger self, and to others who are still finding their passion: follow what makes you curious, and build the kind of community where others can do the same. Approaching graduation, I will carry three things mathematics has taught me—rigorous problem-solving, resilience, and the ability to build meaningful relationships—and I am confident they will define whatever comes next.